

Mohamed Mahmoud

Technical Lead / Senior Software Engineer

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Professional Summary

High-impact **Technical Lead & Senior Software Engineer** with over 7 years of experience architecting, scaling, and engineering mission-critical backend architectures, cross-platform mobile applications. Proven expert in building fault-tolerant, polyglot microservices (Python/FastAPI, PHP/Laravel, Node.js) and multi-tenant SaaS platforms tailored for B2B/B2C. Deeply skilled in advanced design patterns (Factory, Circuit Breaker, Adapter, Builder, Decorator), event-driven architectures (EDA), and complex financial ledgers. Demonstrated leadership in bridging highly concurrent backend aggregation services with decoupled Next.js multi-brand frontends and React Native mobile apps, optimizing system reliability for thousands of active users.

Technical Skills

- **Backend & APIs:** Python, FastAPI, PHP, Laravel 8+, Node.js, REST APIs
- **System Architecture:** Polyglot Microservices, Multi-Tenant SaaS, B2B/B2C Systems, Event-Driven Architecture (EDA)
- **Design Patterns:** Factory, Circuit Breaker, Unified Schema, Adapter, Strategy, Facade, Decorator, Builder
- **Frontend:** Next.js, React.js, Redux, Axios, Ver-
- cel Edge Middleware
- **Mobile:** React Native, Expo, React (RTK Query), (EAS for deployment)
- **Databases & State:** MySQL (ACID), Redis (State Machines), Meilisearch (Inverted Index), MongoDB
- **Infrastructure:** Cloudflare, Nginx, AWS, S3, Docker, CI/CD pipelines

Professional Experience

Technical Lead (Oct 2024 – Present) & Senior Software Engineer (Oct 2022 – Oct 2024)
Riyadh, KSA (Remote) / Oct 2022 – Present
Tripklik

- **Polyglot Distributed Architecture:** Architected an enterprise-grade travel booking ecosystem utilizing **FastAPI (Python)** for high-concurrency flight/hotel search aggregation and **Laravel (PHP)** for core transactional booking and B2B financial ledgers.
- **FastAPI Aggregation & Flight Fulfillment:** Engineered the travel provider integration engine using a **Factory Pattern** and **Unified Schema Mapper** to standardize complex payloads across multiple third-party suppliers (Amadeus, Hotelbeds, Flyadeal).
- **Enterprise API Resilience:** Built an advanced API Resilience Overlay incorporating **Circuit Breakers**, provider-aware adaptive throttling, and exponential backoff to protect internal infrastructure from external supplier downtimes.
- **Financial Core & B2B Accounting:** Designed a robust financial engine in Laravel featuring a dynamic multi-currency conversion system (Supplier → Base → Client) and a **B2B Pocket Management** system utilizing ACID-compliant MySQL ledgers and a Payment Gateway Factory (Stripe, PayPal, B2B wallets).
- **Advanced State Management & EDA:** Implemented a shared **Redis State Machine** bridging ephemeral session data between Python microservices and the PHP booking engine. Orchestrated an **Event-Driven Architecture (EDA)** with background queues to decouple notification services and trigger cross-service auto-booking execution without blocking the user journey.

- **Next.js Multi-Tenant Frontend:** Spearheaded a decoupled multi-brand architecture deploying via a single common branch on Vercel. Utilized Edge Middleware for dynamic subdomain routing, environment-driven **S3 CSS asset injection**, and centralized Axios interceptors for global state, auth, and header management.
- **High-Speed Autocomplete:** Integrated Meilisearch as an inverted index service via direct Nginx routing, drastically reducing database load and delivering ultra-fast autocomplete experiences.
- **Mobile Multi-Branding (Distinia, Tripklik):** Developed a React Native architecture utilizing runtime environment injections to launch distinct, white-labeled client mobile builds from a singular code-base.

Senior Software Engineer

Cairo, Egypt / Feb 2019 – May 2022

Ivas

- **Enterprise Telecom Integrations:** Directed the core backend development and infrastructure setup for large-scale Ringback Tone delivery networks deployed across major telecom operators in the GCC region (KSA, UAE, Kuwait).
- **Architecture Evolution:** Shifted production backends from standard MVC monolithic designs to robust, clean layered architectures using Laravel and MongoDB to facilitate seamless integration flows.
- **Team Mentorship:** Led backend squads in adopting best practices for API security, performance profiling, and maintainable code delivery.

Back-End Developer

Cairo, Egypt / Jun 2018 – Oct 2018

Grand

- **Developed RESTful APIs** serving mobile and web applications with focus on performance and security.
- **Collaborated with frontend teams** to define API contracts and ensure optimal data flow.
- **Managed client communications** providing technical guidance and project updates.

Full-Stack Developer

Cairo, Egypt / Jan 2018 – May 2018

LoadServ For Integrated Business Solution

- **Contributed to team projects** focusing on modern web technologies and best practices.
- **Solved complex technical challenges** in both frontend and backend development.
- **Participated in code reviews** and knowledge sharing sessions.

Education

Bachelor of Science in Computer Science

Cairo, Egypt / 2014 – 2018

Helwan University

- **Graduation Project:** Face Recognition child search platform built with Python, Flask, OpenCV, and Firebase.